PSORAISIS 2

DR EMAN GOMAA

CONSULTANT DERMATOLOGY

VARIENTS OF PSORAISIS

4 major	7 site	2	8 other
Ps vulgaris Guttate ps Erythrodermic Pustular ps	Nail ps Ps arthritis Scalp ps Palmoplanter Mucosal ps Flexural ps Penile ps	Childhood ps Unstable ps	Ruboid Elephantine Ostraceous Digital Interdigetal Verrucous Lichenoid follicular

Pustular ps

generalized

- Acute GPP(vonzumbush)
- GPP in pregnancy
- Circinate ,annular ,linear
- Exanthamatus
- Infantil
- Localized not hand ,feet

localized

- Palmoplanter pustulosis
- Acrodermatitis continue of halopoe

4 • In generalized pustular psoriasis, infiltration of neutrophils dominates the histologic picture, explaining the bright erythema and sterile pustules

- It is an unusual manifestation of psoriasis, and triggering factors include
  - pregnancy,
  - rapid tapering of corticosteroids (or other systemic therapies),
  - hypocalcemia,
  - infections
  - topical irritants.

5 ACUTE GPP (VONZUMBACH)

- Burning pain in skin
- Fever , malaise .
- Sheets of erythema studded with pustules in affected unaffected skin .affect mainly flexural area .
- Nail thickening , onycholysis
- Geographical tongue
- complication
  - Erythroderma , hypoalbuminemia , hypocalcemia , oligemia ,
  - Liver , kidney , joint affection .

  Leukocytosis , increase ESR ,decrease albumin , zinc , ca

• Pp of pregnancy • Impetigo herptiform	• Circinat ,annular ,linear	• exanthametus	• infantile	• Localized except palm ,sole

• Last trimester
• Associated with hypocalcemia
• placental insufficiency leading to stillbirth,neonatal death or fetal abnormalities.
	• Pustules appear peripherally on the advancing edge, dry and leave a trailing fringe of scales
	• DD:
	  • Sneddon wilkkinson
	  • IGA pemohigus
	  • LABD
		• Acute self limited
		• After infection , lithium
		• No systemic symptom
		• DD :AGEP
		(ampicillin , face , flexure, after 24 h from tt , eseanophilis)
			• 2-10yrs
			• Male:female 3;2
			• Benign
			• no systemic
			• sponataneus remission
				• Pustules appear within or edge of ps plaque
				• Unstable phase

7


8


A


B


A
B
C
D

9

10

LOCALIZED PUSTULAR PS

Palmoplanter pustulosis	Acrodermatitis continue of hallopeau
• Bilateral symmetrical sterile pustules • With yellow brownish macules • Sapho syndrome • Triggering by :focal infection , smoking , stress • lithium • Chronic course resisite to treatment • DD: vesiculoboiious tinea , 2ry infected eczema	• Pustules on distal portion of finger or toe • Nail dystrophy ,shedding , shedding • Osteolysis on distal phalynex • Geographical tongue(annulus migrans) • DD: • herpetic whitlow • staph infection • contact dermatitis

Fig. 8.10 Pustulosis of the palms and soles. Multiple sterile papules are admixed with yellow–brown macules on the palm.

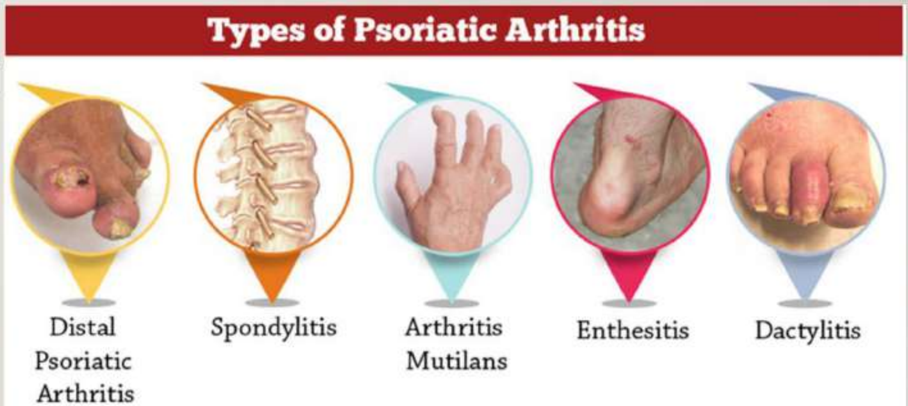
Fig. 8.11 Acrodermatitis continua of Hallopeau. Erythema and slight scale of the distal digit, pustules within the nail bed, and partial shedding of the nail plate.

NAIL PSORAISIS	
5% isolated , 40%-45% skin ps , 80%90% ps arthritis . Finger nail >toe nail	
Nail matrix (PDRL)	**Nail bed**
pitting	hyperkeratosis
leukonychea	onycholysis
Red spot	Oil drops
Plat dyetrophy	Splinter hge
Beaus line	


A. Nail matrix psoriasis
Pitting, leukonychia
Leukonychia
Red macules in lunula
Crumbling
Trachyonychia
B. Nail bed psoriasis
Splinter hemorrhages and onycholysis
Hyperkeratosis and splinter hemorrhages
Salmon patch or oil spot dyschromia
Onycholysis and salmon patch dyschromia

15 PSORAITIC ARTHRITIS

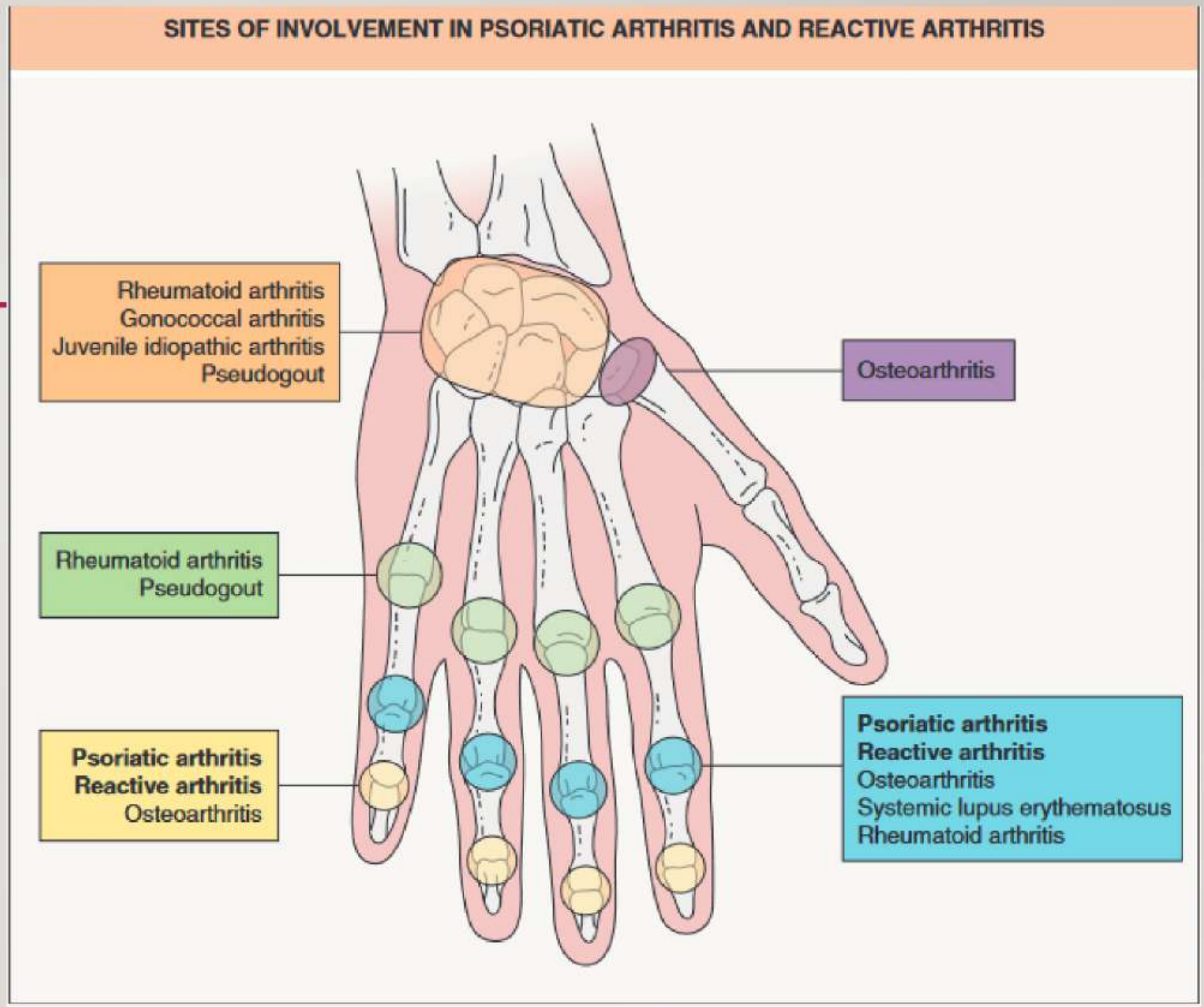
- PSA is 5-30% of skin ps
- Mostly associated with nail ps gluteal cleft , scalp .
- Five classical type(moll &wright)
  1. Mono or asymmetric oligoarthritis (DIP ,PIP , band
  2. DIP
  3. RA like presentation
  4. Arthritis mutilans
  5. Spondylitis , sacroilitis



16

FIVE MAJOR TYPES OF PSORIATIC ARTHRITIS	
Type	**Clinical findings**
Mono- and asymmetric oligoarthritis (most common type)	• Inflammation of the DIP and PIP joints of the hands and feet (Fig. 8.15) • Involvement of the PIP or both the DIP and PIP joints of a single digit can result in the classic "sausage" digit (Fig. 8.16) • MCP joint involvement unusual (unlike RA) • May be accompanied by inflammation of larger joints
Arthritis of the distal interphalangeal joints	• Exclusive involvement of the DIP joints • Classic but uncommon presentation • The joints may become fixed in a flexed position
Rheumatoid arthritis-like presentation (difficult to distinguish from RA clinically*)	• *Symmetric* polyarthritis that involves small and medium-sized joints, in particular the PIP, MCP, wrist, ankle and elbow • Most patients are seronegative for RF
Arthritis mutilans** (least common type)	• Severe, rapidly progressive joint inflammation that results in destruction of the joints and permanent deformity • Digits become shorter, wider and softer to palpation because of osteolysis and a telescoping phenomenon
Spondylitis and sacroiliitis (also have peripheral joint involvement)	• Spondylitis resembles that seen in ankylosing spondylitis, with axial arthritis as well as involvement of the knees and sacroiliac joints • Patients are often HLA-B27-positive and may have associated inflammatory bowel disease and/or uveitis

*Whether some seropositive patients have an overlap of the two disorders is possible.
**Also occurs in multicentric reticulohistiocytosis.


SITES OF INVOLVEMENT IN PSORIATIC ARTHRITIS AND REACTIVE ARTHRITIS
Rheumatoid arthritis
Gonococcal arthritis
Juvenile idiopathic arthritis
Pseudogout
Osteoarthritis
Rheumatoid arthritis
Pseudogout
Psoriatic arthritis
Reactive arthritis
Osteoarthritis
Psoriatic arthritis
Reactive arthritis
Osteoarthritis
Systemic lupus erythematosus
Rheumatoid arthritis

scalp	mucosal	palmoplanter	flexural	Penile ps

• Common may be the only site
• thick adherent scale
• Pattern
  • Discret plaque
  • Diffuse
  • Marginal
  • Dosnt cause alopecia
	• Oral
	Annulus migrans (geographical tongue)
	In acrodermatitis continue of hallopeau
	GPP
	• ocular
		1. Typical scaly patch
		2. Less well defined plaque
		3. Pp pustulosis
		4. mixed
			• Shiny , sharply demarcated thin plaque
			• No scale
			• Older adult
				Typical ps plaque
				Scaly on circumcised lesion
				Non scaly on non circumcised penis

Fig. 8.12 Scalp psoriasis with extension onto the neck. Note the involvement of the external auditory canal.

20

21

Childhood ps	Unstable ps	other

• 40% of ps cases
• +ve family history
• Triggering factor (infection ,trauma)
• Most common :
  • Plaque
  • Napkin
  • Guttat
  • Facial , flexur
  • No pustular
  • noerythrodermic
	• Risk fator:
	  • Hypocalcemia ,infection , tar , uv , sever emotional stress
		• Ruboid :cone shape
		• Elephantine :thick scaling
		• Ostraceous :ring concave
		• Digital
		• Inter digital
		• Verrucous
		• Lichenoid
		• follicular

23

Disorder associated with psoraisis

Skin disease

- Sd
- Lsc
- Skin cancer :with puva , cyclosporin
- Decrease infection
- Decrease allergic skin disease
- Antip200 pemphiguse

Systemic disease

1. Metabolic syndrome
2. Cardiovasculardisease
3. IBD :HLA27
4. Steatohepatitis
5. Uveitis
6. Cancer :lymphoma
7. psychological

Similar to ps but distenect disease

1. Sneddon wilkinson disease
2. Reiters (reactive arthritis)
3. ILVEN

- Sneddon Wilkinson disease:
    - Annular pusules in flexur
    - Non immune mediated –ve IF
    - IgA paraproteinemia

26 REITERS DISEASE

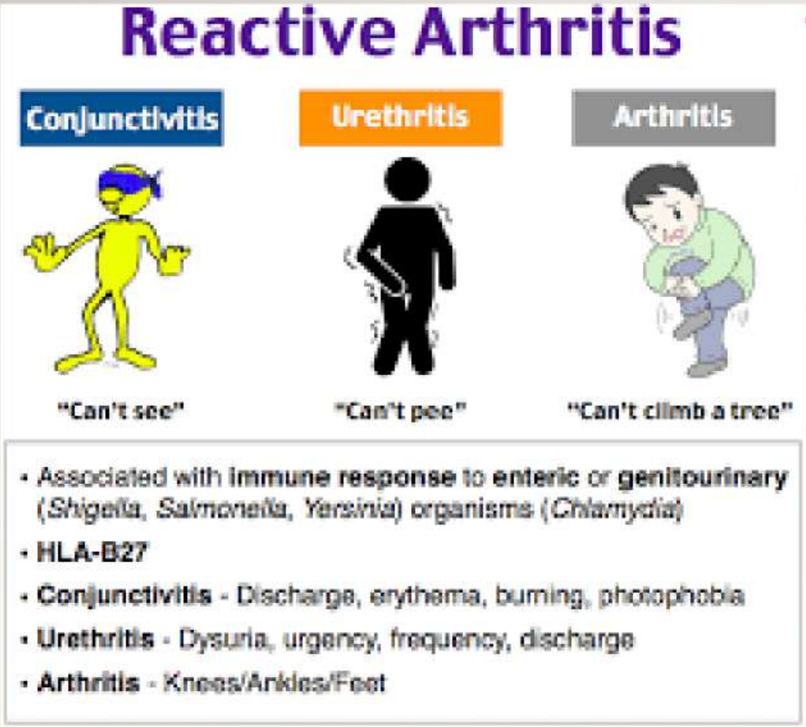


Fig. 8.17 Reactive arthritis (formerly Reiter disease). A, B Plantar lesions of keratoderma blennorrhagicum. The amount of scale can vary and when sterile pustules form, they are often admixed with yellow–brown macules. **C** Papulosquamous lesions on the glans and shaft of the penis. *B, Courtesy, Eugene Mirrer, MD.*

29 ASSESSMENT OF PSORIASIS SEVERITY

a) Psoriasis area and severity index (PASI) score:

• Divide the body into 4 regions: head and neck (0.1 of body surface), upper limbs (0.2 of body surface), trunk (0.3 of body surface) and lower limbs (0.4 of body surface).

• In each region, assess:

A Severity of erythema, scaling and induration in 0 - 4 scale (0 = none; 1 = slight; 2 = moderate; 3 = severe; 4 = very severe).

A Area of involvement in 0 - 6 scale (0 = none; 1 = <10%; 2 = 10-30%; 3 = 30 - 50%; 4 = 50 - 70%; 5 = 70 - 90%; 6 = 90 - 100%).

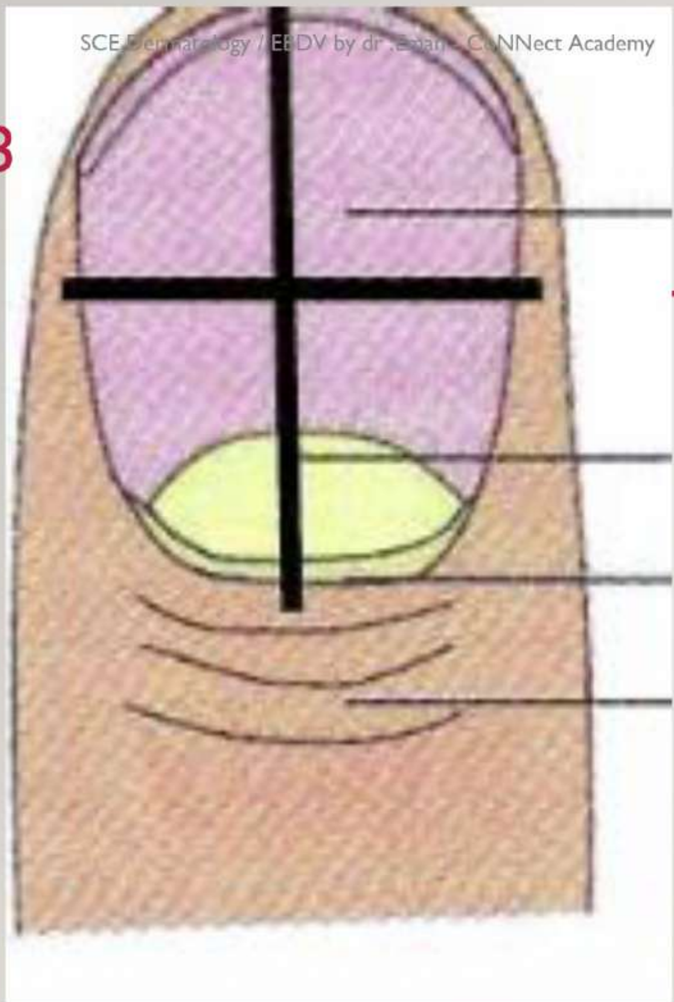
CALCULATION OF THE PSORIASIS AREA AND SEVERITY INDEX				
Severity of psoriatic lesions				
[0, none; 1, slight; 2, moderate; 3, severe; 4, very severe]				
	Head	***Trunk***	***Upper limbs***	***Lower limbs***
Erythema	0 to 4	0 to 4	0 to 4	0 to 4
Induration	0 to 4	0 to 4	0 to 4	0 to 4
Scaling	0 to 4	0 to 4	0 to 4	0 to 4
Total score = ❶	Sum of the above	Sum of the above	Sum of the above	Sum of the above
Area of psoriatic involvement				
[0, none; 1, <10%; 2, 10 to <30%; 3, 30 to <50%; 4, 50 to <70%; 5, 70 to <90%; 6, 90–100%]				
Degree of involvement = ❷	0 to 6	0 to 6	0 to 6	0 to 6
Multiply ❶ × ❷	❶ × ❷	❶ × ❷	❶ × ❷	❶ × ❷
Correction factor for area of involvement = ❸	0.10	0.30	0.20	0.40
❶ × ❷ × ❸	**A**	**B**	**C**	**D**
A + **B** + **C** + **D** = total PASI				

	Head & neck	*Upper limbs*	*Trunk*	*Lower limbs*
Erythema	0 – 4	0 – 4	0 – 4	0 – 4
Induration	0 – 4	0 – 4	0 – 4	0 – 4
Scaling	0 – 4	0 – 4	0 – 4	0 – 4
A (sum of 3)	**0 – 12**	**0 – 12**	**0 – 12**	**0 – 12**
B (area)	**0 – 6**	**0 – 6**	**0 – 6**	**0 – 6**
C (% of body surface)	**0.1**	**0.2**	**0.3**	**0.4**
A x B x C	**H**	**U**	**T**	**L**
PASI score = H+U+T+L (range 0 – 72)				

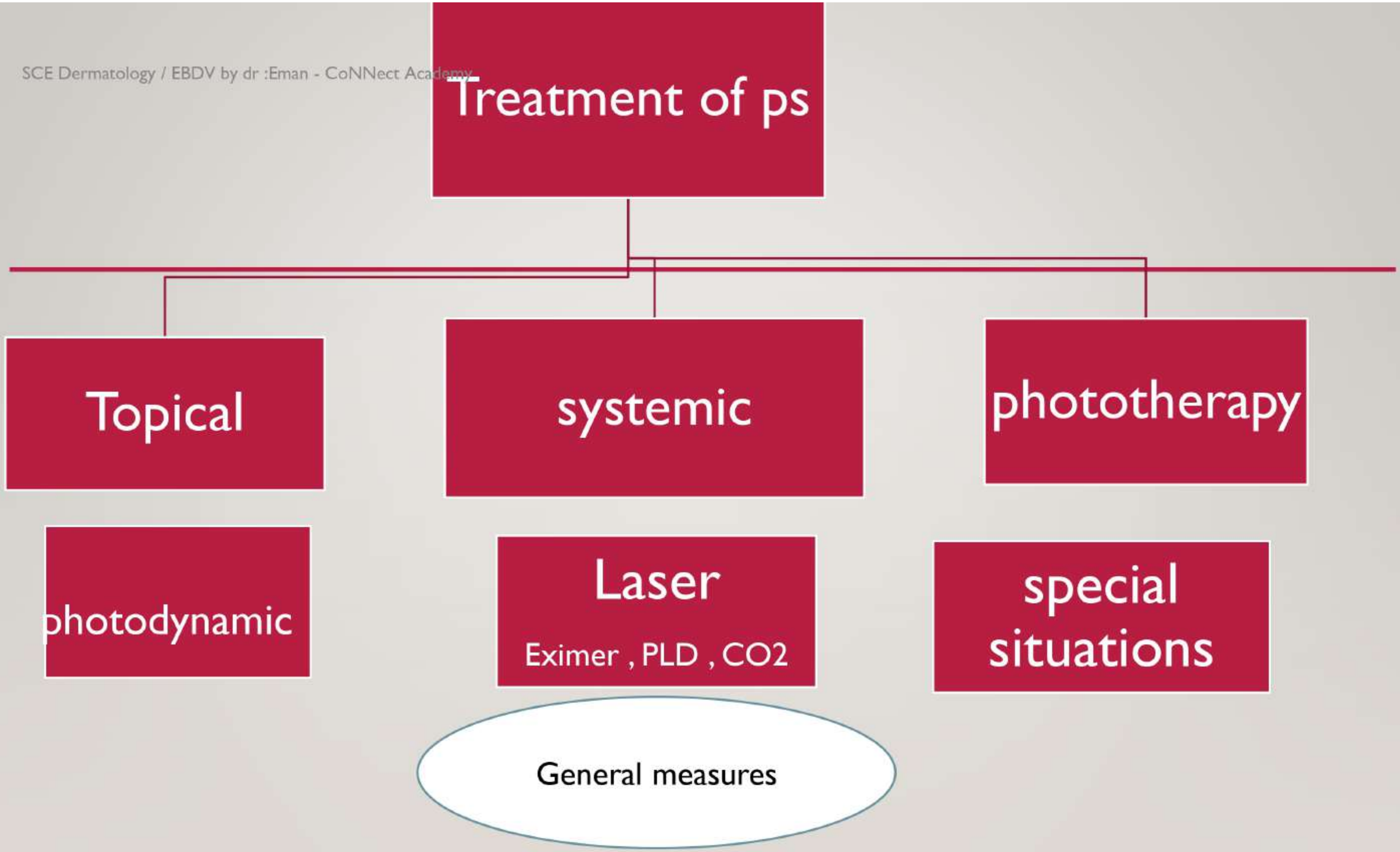
32 NAIL PSORIASIS SEVERITY INDEX (NAPSI) SCORE:

- .Divide the nail with imaginary horizontal and longitudinal lines into 4 quadrants.

• Each nail is given a score in each quadrant for:
A Nail bed psoriasis (onycholysis, splinter hemorrhages, subungual hyperkeratosis, oil drop "salmon patch")
A Nail matrix psoriasis (pitting, leukonychia red spots in the lunula, crumbling)
0 = none
1 = present in 1 quadrant of the nail
2 = present in 2 quadrants of the nail
3 = present in 3 quadrants of the nail
4 = present in 4 quadrants of the n



- each nail gets a matrix and a bed score, the total range is 0 - 8.
  The total NAPSI score = sum of all the nails (0 - 80 for fingernails;
  or 0 - 160 if toenails are included).


Treatment of ps
Topical
systemic
phototherapy
photodynamic
Laser
Eximer , PLD , CO2
special situations
General measures

TOPICAL TREATMENT

1. STEROID
2. TARULIMUS
3. VIT D3
4. RETINOID
5. SALICYLIC ACID
6. COAL TAR
7. ANTHRALINE

SYSTEMIC TREATMENT

1. METHOTREXAT
2. ACETRITINE
3. CYCLOSPORINE
4. BIOLOGICS

Topical treatment

steroid	Calcineurin inhibitor	Vit D3 analogue
• 1st line in mild to moderat ps as monotherapy • Sever ps as combination therapy • Antiinflam. Antiproliferative , immunosuppressive , vc S.E: atrophy telangectasia , strea , rosacea ,glaucoma ,cataract tachyphlaxisis Contraindication: bacterial , viral infection • Category c	• Tacrolimus 1%, pimecrolemus 1% • Thin skin (face , intertregenous area • Decrease t cell activation • SE: burning , itching , malignancy with animals • No contaraindication • Categoty c	• Calcipotriol 50 mcg /g • Maximum dose • 100g/ week • Mech:dcrease proliferation , normalize differentiation • S.E : irretaion ,reversable hypercalcemia , decrease parathyroid H • Contraindication:large area , abnormality in bon , ca , renal inssuffeciency , pregnancy , lactation

retinoids	Salicylic acid	Coal tar
• 0.05% and 0.1% cream and gel - applied once daily - maximum area to be treated is 10- 20% of body surface. Mechanism: • Decreases keratinocyte hyperproliferation • Normalizes differentiation • Decreases expression of inflammatory cytokines. Side effects • Local irritation in lesional and perilesional skin • Teratogenicity (pregnancy category X) • Photosensitivity Contraindications Pregnancy, lactation and patients <18 years old Notes • Combination with topical corticosteroids (helps reduce irritation).	• Salicylic acid 5-10% - applied once daily (2 -3 times per week Mechanism: • Reduces keratinocyte-to-keratinocyte binding AND reduces the pH of the stratum comeum —>reduced scaling and softening of ps plaque SE: • Systemic absorption and salicylate toxicity Contraindications • Avoid use with other oral salicylate drugs (risk of systemic toxicity) • Better to avoid in children • Safe for localized psoriasis in pregnancy.	• a distillation product from coal, Mechanism • Anti-mitotic (suppresses DNA synthesis), anti-pruritic and anti-inflammatory. Side effects • Not cosmetically acceptable (staining of clothes / odor), irritant contact dermatitis, folliculitis, and photosensitivity to UVA. • Carcinogenic in animals (no evidence in humans Contraindications • Use with caution during pregnancy, lactation and in children. Notes • Goeckerman regimen (a combination of crude coal tar with UV light)

ANTHRALINE

• "Short-contact therapy " starting at 1% concentration
(20-30 minutes) with increasing concentration over time
as tolerated up to 2% applied twice daily.
Mechanism
• Inhibits T-lymphocyte activation.
• Normalizes keratinocyte differentiation.
SE • Staining of skin, nails and clothing.
• Skin irritation
Contraindications • Applied with caution to face and intertriginous areas
(severe skin irritation).
Notes • Ingram regimen (a combination of dithranol with UV)

39

Major histocompatibility complex (MHC) Class I molecules:

A. Are inducible on keratinocytes
B. Complexed with antigen trigger cytotoxic T cells
C. Are recognized by receptors on CD4+ T cells
D. Bear peptides derived from pathogens taken up into vesicles
E. All of the above

Correct Answer: B

MHC Class I molecules are present on all nucleated cells. They are recognized by receptors on surfaces of CD8+ T cells, and, when complexed with antigen, trigger cytotoxic T cells. The other statements apply to MHC Class II molecules.

40

Psoriatic arthritis is most commonly associated with which HLA?

A. HLA-B27
B. HLA-Cw6
C. HLA-Aw19
D. HLA-Bw35
E. None of these options are correct

Correct Answer: A

HLA-B27 is associated with an increase in psoriatic arthritis as well as pustular psoriasis and acrodermatitis continua of Hallopeau.

41

Which antibody can bind the FcER1 portion of mast cells, basophils, Langerhans cells, dermal dendritic cells?

A. IgA
B. IgD
C. IgE
D. IgG

Correct Answer: C

IgE is an anaphylactic antibody that is involved in nearly all immediate allergic and anaphylactic type reactions and commonly seen in elevated levels in patients with atopic dermatitis. Mast cells, basophils, Langerhans cells, dermal dendritic cells as well as monocytes from atopic individuals all express high-affinity FcERI receptor which can bind IgE. More recently, it became clear that can bind monomeric IgE via the high-affinity FcRI IgG is the antibody that can cross the placenta and the most common antibody found in circulation. IgA is found in mucous membrane secretions and is able to agglutinate antigens and activate the alternate but not the classic complement pathway. IgD is not found in circulation other than in hyper-IgD syndrome, an autosomal recessive disorder caused by mutations in the mevalonate kinase gene. A significant elevation of serum IgD is seen in 95% of these patients. IgM is the antibody produced in the early stages of antibody responses. It is a pentamer which can agglutinate antigen and active the classic complement pathway.


thankyou
for your purchase